

Shaan Patel

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OBJECTIVE

Ambitious third-year Computer Engineering student specializing in computing hardware, VLSI design, and emerging architectures. Seeking a Fall/Spring internship in computer architecture, embedded systems, or hardware design to apply strong programming skills and an end-to-end approach to creating complex systems.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science - GPA: 3.57; Master of Science in ECE

August 2023 – May 2027; August 2027 – May 2028

Major in Computer Engineering, Minor in Applications of Artificial Intelligence and Machine Learning

- Relevant Coursework: Data Structures and Algorithms, Embedded Systems Design, Advanced Microarchitecture Design
- Leadership: Captain of GT Ramblin' Raas, GTXR Executive Board, SiliconJackets Digital Design Team Lead

TECHNICAL SKILLS

Programming: C++, Verilog, C, VHDL, RISC-V, SystemVerilog, OpenGL, Python, Java, JavaScript, HTML/CSS, SQL, LaTeX

Platforms: Linux (Ubuntu, Slackware, Debian), Red Hat (OpenStack, CloudForms), OpenVMS, Solaris, Windows, MacOS

Hardware: FPGA, ESP32, IC Breadboard Circuits, Oscilloscope, Arduino, Raspberry Pi, ARM MBED, LaunchPad, 3D Printing

Frameworks: React, SFML, CUDA, PyTorch, AWS (S3, DynamoDB, EC2), Express, MongoDB, Pandas, Node.js, Kubernetes

Software: KiCAD, Altium, Cadence, Intel Altera Quartus II, LTspice, NI LabVIEW, gRPC, TCP/UDP Tunnels, Blender, Autodesk Fusion 360/AutoCAD, SolidWorks, Onshape, Visual Studio Code, GitHub, Wireshark, Adobe Creative Suite, Inkscape, Slack

Communication: Design proposals, technical reports, instruction manuals, presentations, email correspondence, training materials

EXPERIENCE

AeroCore-V – Hardware-Accelerated SoC for Autonomous UAVs

Atlanta, GA

SiliconJackets Project Lead

August 2025 – Present

- Designed an open-source RISC-V SoC compliant with RV32I ISA standards in SystemVerilog with custom ALU extensions for single-cycle PID control math, tailored for low-latency flight adjustments on a DE10 validated with Intel Quartus Prime
- Integrated a direct-mapped L1 cache for handling high-frequency I2C/SPI GPS and IMU data streams from an ESP32, while concurrently developing an L2 cache controller with an LRU replacement policy
- Developing a virtual memory system with Page Table Walker and TLB to isolate flight routines and protect from page faults, utilizing Github Actions CI/CD pipelines to automate regression testing and cycle-accurate simulations
- Prototyped a priority-based thread scheduler to manage UAV flight loops and telemetry tasks, alongside an OpenGL Digital Twin physics simulation in C++ to validate future flight latency reductions

Georgia Tech HIVE Makerspace

Atlanta, GA

Student Leader and Volunteer

August 2024 – Present

- Instruct ~40 students per week on advanced design and fabrication workflows, including 3D CAD/slicing, circuit analysis with oscilloscopes, laser cutter operation, PCB design and manufacturing (taking a blank FR4 sheet to an operational PCB)
- Deployed an open-source multi-device print manager to automate jobs on a FIFO basis, increasing the operational uptime of the free-to-use 12-unit Bambu Lab 3D print farm for the Georgia Tech student body
- Facilitate workshops and created training resources that contributed to a 20% increase in student participation
- Architecting a full-stack inventory management system using an AWS RDS backend to track materials in the cloud with S3 and DynamoDB to enable a student electronics rental program, aiming to reduce annual student costs

High-Performance Heterogeneous Computing Infrastructure

Ashburn, VA

BlendFarm Project Leader

June 2022 – July 2025

- Assembled two 2.5kW parallel computing systems from the component level, integrating twelve 12GB GDDR6 GPUs and modifying voltage curves to bypass LHR limiters, achieving a 73% hash rate improvement
- Overcame thermal throttling by designing and CNC-fabricating monolithic aluminum water blocks with Solidworks to cool a 6-GPU array, reducing core temperatures by over 15°C under sustained 100% load
- Scripted a distributed render farm using C++ and SFML to coordinate large ray tracing frame rendering across a network
- Optimized task allocation using TCP/UDP socket tunnels and a custom jitter buffer implementation to ensure reliable data synchronization between the master node and rendering clients for faster computer graphics generation

Academy of Science International Research

Ashburn, VA

Research Collaborator

November 2022 – July 2024

- Synthesized a saline-soluble polymer as a novel alternative to currently used petrochemical-based plastics as a means of preventing marine pollution
- Modeled and fabricated 3D-generated molds using filament and resin printers for synthesizing hydrogels through incubation
- Worked with a tensile tester to measure Young's modulus and SEM electron microscopy to analyze sample composition
- Led collaboration efforts with South Korean research teams for two years to formulate samples and complete testing
- Collected over 150 trials of data and was awarded the Yale Outstanding Project award at RSEF in Spring of 2023